Filipino Adaptation of OmniVoice: Fine-Tuning a Multilingual Text-to-Speech Model Using the Philippine Languages Database

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Abstract

This study investigates Filipino adaptation of OmniVoice, a pretrained massively multilingual zero-shot text-to-speech model, using the Filipino subset of the Philippine Languages Database. Although OmniVoice already includes Filipino, the model reports only 7.71 hours of Filipino in its multilingual training mixture, leaving room to examine whether supervised Filipino fine-tuning improves generated speech. The methodology prepares paired Filipino audio and transcript data, filters read-speech utterances, converts the dataset into OmniVoice JSONL and tokenized WebDataset format, fine-tunes the pretrained checkpoint under Kaggle T4 GPU constraints, and evaluates generated audio against the base model. Training and evaluation runs are tracked in Weights and Biases to support hyperparameter comparison across learning rates. Evaluation uses word error rate for intelligibility, objective speaker similarity for voice cloning quality, and UTMOS for predicted naturalness, with listening checks used to interpret metric changes. The final controlled comparison evaluates the base model against three fine-tuning runs that share the same 5000-step budget and best-development-loss checkpoint policy and differ only in learning rate (5e-6, 1e-5, 2e-5). The 1e-5 checkpoint is the strongest overall, reducing WER from 22.55% to 18.52% while keeping objective speaker similarity slightly above the base model; the 2e-5 checkpoint nearly matches that intelligibility gain but lowers speaker similarity, and the 5e-6 checkpoint preserves speaker similarity best while improving WER only modestly. The results show an intelligibility-versus-speaker-similarity tradeoff governed by learning rate rather than a single-metric win.

1 Introduction

Text-to-speech (TTS) and voice cloning systems generate speech waveforms from text and, in zero-shot voice cloning settings, imitate a target speaker from a short reference recording. Recent neural speech generation models increasingly use large-scale pretrained models, discrete audio tokens, and generative architectures to synthesize natural and speaker-conditioned speech (Wang et al., 2023; Le et al., 2023; Wang et al., 2024; Zhu et al., 2026). These systems are useful for accessibility, education, localization, narration, and language-learning tools. However, practical speech generation quality remains uneven across languages because high-resource languages tend to dominate available datasets and benchmarks.

Filipino is widely used in the Philippines, but open Filipino TTS and voice cloning workflows remain less mature than English-focused systems. OmniVoice is a promising starting point because it is open source, supports more than 600 languages, and includes Filipino with language ID `fil` (Zhu et al., 2026). However, language support does not automatically imply strong performance in pronunciation, intelligibility, naturalness, or speaker similarity. The OmniVoice language table lists 7.71 hours of Filipino in its training mixture, while the Philippine Languages Database (PLD) reports 48:56:36 of Filipino speech (Cajote et al., 2024). This difference motivates an adaptation study using supervised fine-tuning.

1.1 Problem Statement

Open Filipino speech generation systems face two related limitations. First, high-quality voice cloning systems are often optimized around languages with larger training corpora and stronger evaluation coverage. Second, multilingual support in a pretrained model may still be thin for a specific low-resource language, especially when the available language-specific hours are small relative to the full training mixture.

The machine learning problem addressed in this study is:

> Given a pretrained multilingual TTS model and a curated Filipino speech-text dataset, determine whether supervised fine-tuning improves Filipino speech generation compared with the original base checkpoint.

The study evaluates improvement using objective metrics for intelligibility, speaker similarity, and naturalness, then interprets those metrics with generated audio samples. The expected outcome is a reproducible fine-tuning and evaluation workflow for measuring Filipino adaptation of OmniVoice.

2 Related Works

Voice cloning is closely related to text-to-speech, but its goal is more specific: the system must generate the requested text while preserving the voice characteristics of a target speaker. Azzuni and El Saddik (2025) frame modern voice cloning around speaker adaptation, few-shot TTS, zero-shot TTS, and multilingual TTS. This distinction is important for the present study because different settings use different amounts of target data. Speaker adaptation usually updates a model using speaker or language examples, while zero-shot voice cloning tries to imitate a reference voice without updating the model at inference time. This study uses a zero-shot model as its starting point, but the research task itself is supervised adaptation: Filipino text-audio pairs are used to fine-tune the pretrained model.

Recent TTS research has moved from narrowly trained speaker-specific systems toward large pretrained speech generation models. VALL-E helped establish neural codec language modeling for TTS by generating discrete audio codec codes conditioned on text and a short acoustic prompt (Wang et al., 2023). Voicebox extended this direction with text-guided speech infilling and multilingual training, showing that one speech generation model can support several generation and editing tasks (Le et al., 2023). MaskGCT and F5-TTS further reflect the trend toward non-autoregressive generation, where systems aim to produce high-quality speech more efficiently than step-by-step autoregressive decoding (Wang et al., 2024; Chen et al., 2024). Across these works, the shared pattern is clear: modern voice cloning increasingly depends on pretrained generative models, speech tokenizers, and short speaker prompts rather than training a separate model for every speaker.

OmniVoice is the work most directly connected to this project because it combines those trends with broad language coverage. Zhu et al. (2026) present OmniVoice as a massively multilingual zero-shot TTS model with a diffusion language model-style, discrete, non-autoregressive architecture. Its main difference from two-stage systems is that it does not first predict semantic tokens and then convert them into acoustic tokens. Instead, OmniVoice directly maps text to multi-codebook acoustic tokens. The model uses full-codebook random masking during training and initializes from pretrained LLM weights to improve intelligibility. It also uses the Higgs Audio tokenizer to extract 8-codebook acoustic tokens and reconstruct speech waveforms (Zhu et al., 2026; Boson AI, 2025). These design choices make OmniVoice a practical base model for a Filipino adaptation experiment: the model already supports Filipino, but it can still be tested for improvement when exposed to more Filipino paired speech data.

The machine learning method in this project is supervised fine-tuning, which is a form of transfer learning. Transfer learning reuses knowledge learned from a source setting to improve learning in a related target setting (Pan and Yang, 2010). In this study, the source setting is OmniVoice's pretrained multilingual speech generation model, and the target setting is Filipino TTS and voice cloning. The supervised labels are the paired PLD transcripts and audio recordings. Fine-tuning is appropriate here because training a modern TTS model from scratch would require far more data and compute than the project constraints allow, while updating a pretrained checkpoint can adapt the text-to-acoustic-token mapping toward Filipino pronunciation and usage under a Kaggle T4 workflow.

The dataset source is the Philippine Languages Database introduced by Cajote et al. (2024). PLD was developed as a multilingual speech corpus for Philippine spoken languages and covers Filipino, English, Cebuano, Kapampangan, Hiligaynon, Ilokano, Bikolano, Waray, and Tausug. The corpus is relevant because it provides the kind of paired speech and transcript data required for supervised TTS fine-tuning. The Filipino subset also gives this study a concrete gap to investigate: OmniVoice already lists Filipino support, but its reported Filipino hours are much smaller than the Filipino speech available in PLD. The project therefore asks whether Filipino-specific supervised adaptation improves the base model instead of assuming that multilingual support alone is sufficient.

Evaluation in recent zero-shot TTS work usually separates three concerns: intelligibility, speaker preservation, and naturalness. Following Voicebox and OmniVoice, this study uses WER or CER for intelligibility, SIM-o for objective speaker similarity, and UTMOS for predicted naturalness (Zhu et al., 2026; Le et al., 2023). WER and CER are based on edit distance, commonly associated with Levenshtein's insertion, deletion, and substitution operations (Levenshtein, 1966). Because WER depends on the ASR model used to transcribe generated speech, this study also cites Whisper, HuBERT, and Paraformer as relevant ASR backends used in modern speech evaluation pipelines (Radford et al., 2022; Hsu et al., 2021; Gao et al., 2022). Speaker similarity depends on the embedding model, so ECAPA-TDNN and WavLM are included because OmniVoice evaluation uses WavLM-based ECAPA-TDNN speaker embeddings (Desplanques et al., 2020; Chen et al., 2021). UTMOS is used as an automatic MOS predictor for naturalness (Saeki et al., 2022). Together, these metrics support the main comparison of this project: whether fine-tuned OmniVoice produces Filipino speech that is clearer without losing speaker identity or naturalness.

3 Objectives

3.1 General Objective

The general objective is to fine-tune and evaluate a pretrained multilingual TTS model for Filipino speech generation using the Philippine Languages Database.

3.2 Specific Objectives

1. Prepare a clean Filipino read-speech dataset from the Philippine Languages Database.
2. Convert the prepared dataset into the JSONL and tokenized WebDataset format required by OmniVoice.
3. Fine-tune the pretrained `k2-fsa/OmniVoice` checkpoint on Filipino speech using Kaggle T4 GPUs.
4. Track training and evaluation runs in Weights and Biases for hyperparameter comparison across learning rates.
5. Generate Filipino speech samples from both the base and fine-tuned models using matched test prompts and reference voices.
6. Evaluate both models using intelligibility, speaker similarity, and naturalness metrics.
7. Interpret whether Filipino-specific fine-tuning improves generated Filipino speech under limited compute constraints.

4 System Design and Methodology

4.1 System Architecture

The system architecture separates reusable data artifacts, compute-heavy training jobs, and downstream evaluation or delivery outputs.

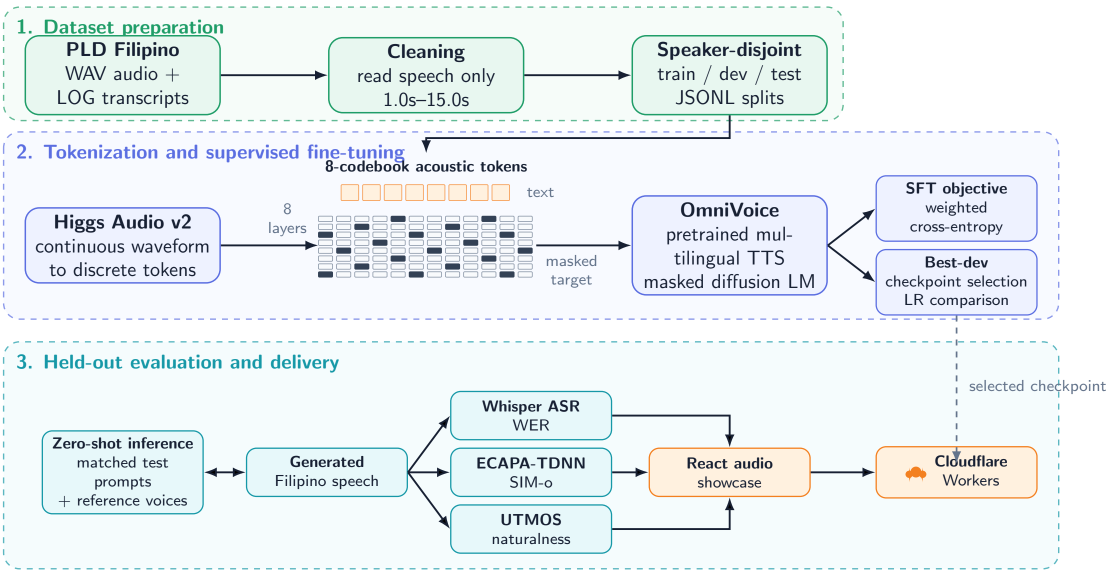

Figure 1
System architecture for Filipino OmniVoice fine-tuning, evaluation, and public showcase delivery.

4.2 Dataset Preparation

The primary dataset is the UP-DSP Philippine Languages Database released through Mozilla Data Collective. The corpus contains WAV audio recordings and LOG transcript files. The

Filipino subset statistics from the PLD paper are shown in Table 1. The Filipino subset is large enough for a supervised adaptation experiment and contains substantially more Filipino speech than the 7.71 Filipino hours listed in OmniVoice's original multilingual training mixture.

Table 1

Filipino subset statistics from the PLD paper (Cajote et al., 2024).

Field	Value
Language	Filipino
Language ID	`fil`
Speakers	135
Utterances	52,879
Duration	48:56:36
Average utterance length	3.5796 seconds

Figure 2

PLD language statistics table excerpt.

Language	Gender	Speaker Count	Utterance Count	Audio Duration Total (h:m:s)	Audio Duration Average (s)	Tokens Total	Tokens Unique
Bikolano (bik)	F	121	39,260	60:55:58	5.5873	321,721	16,049
	M	85	27,684	40:17:36	5.2397	206,642	14,631
	all	206	66,944	101:13:35	5.4436	528,363	17,005
Cebuano (ceb)	F	86	34,956	35:47:01	3.6852	144,882	8,026
	M	66	27,477	27:44:58	3.6357	114,563	7,267
	all	152	62,433	63:31:59	3.6634	259,445	6,844
English (eng)	F	23	3,156	4:31:30	5.1617	29,376	4,363
	M	7	888	1:01:40	4.1675	8,050	1,483
	all	30	4,044	5:33:11	4.9434	37,426	4,729
Filipino (fil)	F	79	30,617	31:43:55	3.7311	205,346	10,861
	M	56	22,262	20:50:48	3.3712	138,088	7,994
	all	135	52,879	48:56:36	3.5796	343,434	11,481
Hiligaynon (hil)	F	48	17,079	21:49:43	4.6012	99,087	5,397
	M	43	14,908	19:21:58	4.6766	84,676	4,906
	all	91	31,987	41:11:42	4.6363	183,763	5,767
Ilokano (ilo)	F	64	15,429	25:46:37	6.0145	131,316	11,603
	M	60	14,513	25:44:43	6.3862	130,500	11,642
	all	124	29,942	51:31:20	6.1947	261,816	13,270
Kapampangan (pam)	F	104	35,024	49:42:02	5.1086	225,595	12,827
	M	83	26,926	40:37:22	5.4313	176,947	12,629
	all	187	61,950	90:19:25	5.2488	402,542	14,221
Pangasinan (pag)	F	12	3,959	6:01:36	5.4802	24,819	4,302
	M	6	1,945	3:07:00	5.7687	11,698	3,148
	all	18	5,904	9:08:36	5.5753	36,517	4,773
Tausug (tsg)	F	4	1,185	1:43:09	5.2236	12,684	2,023
	M	9	2,103	3:06:34	5.3233	7,279	1,536
	all	13	3,288	4:49:45	5.2874	19,963	2,376
Waray-Waray (war)	F	48	15,337	22:51:12	5.3643	94,764	6,071
	M	26	8,500	12:04:10	5.1118	52,704	5,518
	all	74	23,837	34:55:23	5.2743	147,468	6,291
Total	-	**1,030**	**343,208**	**454:49:43**	**4.7708**	**2,220,737**	-

Table 2: Summary statistics for the Philippine Languages Database. Below the each Philippine language name is its language ID in parenthesis, based from the ISO 639-3 standard, as published in Ethnologue (Eberhard et al., 2024) Note that the total token and unique token counts do not include yet the transcripts from the spontaneous speech part as this part of the corpora is still being transcribed.

Only Filipino read speech is used in the initial experiment. Read speech is preferred because each utterance is expected to match a prepared transcript, which is important for supervised TTS fine-tuning. Spontaneous speech is excluded because a speaker's response may not match the displayed prompt, increasing transcript-audio mismatch risk.

The data preparation script parses PLD files and writes OmniVoice-compatible JSONL rows. Each row contains an utterance ID, relative audio path, transcript, and language ID.

```
{"id":"0000.110816.021250.0001",
 "audio_path":"wavs/0000/0000.110816.021250.0001.wav",
 "text":"manugang",
 "language_id":"fil"}
```

The cleaned local package is structured as follows:

```
data/pld_filipino_clean/
  wavs/
  train.jsonl
  dev.jsonl
  test.jsonl
```

The current split is shown in Table 2. Speakers are disjoint across train, development, and test splits to reduce speaker leakage between training and evaluation.

Table 2
Prepared Filipino dataset split.

Split	Samples	Percent	Hours	Speakers
Train	33,448	79.12%	31.902	103
Development	4,506	10.66%	3.705	13
Test	4,322	10.22%	4.165	13
Total	42,276	100.00%	39.772	129

The cleaning process keeps read Filipino speech, removes spontaneous prompt sources, excludes transcripts containing parentheses or digits, removes unreadable or zero-byte WAV files, keeps clips between 1.0 and 15.0 seconds, and applies light transcript normalization. The filtering is intentionally conservative because TTS fine-tuning is sensitive to noisy text-audio alignment.

The cleaned corpus spans varied prompt domains. The largest read-speech sources are news sentences (4,330 clips), medical sentences (2,397), time expressions (1,973), common expressions (1,941), short stories (1,904), greetings (1,708), interrogatives (1,634), and Filipino proverbs or *salawikain* (1,620). Many of the remaining sources are isolated-word prompts such as names, local cuisines, landmarks, and ordinals. This composition explains the short 3.4-second average clip length and matters when interpreting results: the model sees many single-word targets, which limits how much sentence-level prosody it can learn from this corpus. The 31.9 training hours are also roughly four times the 7.71 Filipino hours in OmniVoice's original training mixture, which is the quantitative basis for expecting fine-tuning to help.

4.3 Model Selection and Adaptation

The selected model is `k2-fsa/OmniVoice`. OmniVoice was chosen because it is open source, supports Filipino, provides pretrained checkpoints, supports voice cloning, and includes official fine-tuning and evaluation scripts. The study uses supervised fine-tuning and transfer learning. The base OmniVoice model has already learned multilingual speech generation from a large corpus, while the Filipino PLD subset provides paired Filipino text and audio examples for adapting the model's text-to-acoustic-token mapping.

Figure 3
OmniVoice paper first page.

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OmniVoice: Towards Omnilingual Zero-Shot Text-to-Speech with Diffusion Language Models

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Abstract

We present OmniVoice, a massively multilingual zero-shot text-to-speech (TTS) model that scales to over 600 languages. At its core is a novel diffusion language model-style discrete non-autoregressive (NAR) architecture. Unlike conventional discrete NAR models that suffer from performance bottlenecks in complex two-stage (text-to-semantic-to-acoustic) pipelines, OmniVoice directly maps text to multi-codebook acoustic tokens. This simplified approach is facilitated by two key technical innovations: (1) a full-codebook random masking strategy for efficient training, and (2) initialization from a pre-trained LLM to ensure superior intelligibility. By leveraging a 581k-hour multilingual dataset curated entirely from open-source data, OmniVoice achieves the broadest language coverage to date and delivers state-of-the-art performance across Chinese, English, and diverse multilingual benchmarks. Our code and pre-trained models are publicly available[1].

1 Introduction

Zero-shot text-to-speech (TTS) models trained on large-scale datasets have demonstrated a remarkable ability to generate high-quality speech conditioned on only a few seconds of reference audio [1, 2]. Despite these advances, most existing models support only a limited set of languages, often leaving hundreds of low-resource languages behind. Expanding language coverage is not merely a technical challenge but also a crucial step toward extending speech technologies to languages across the globe. To address this gap, we aim to develop a massively multilingual zero-shot TTS model supporting hundreds of languages. Realizing such extensive scaling requires a TTS architecture with exceptional capacity and robustness to handle diverse linguistic patterns.

In the pursuit of such scalable and high-quality TTS, current research primarily adheres to two paradigms: autoregressive (AR) [3, 4, 5, 6, 7, 8, 9, 10, 11] and non-autoregressive (NAR) models [12, 13, 14]. NAR models offer advantages in both inference speed (via parallel decoding) and robustness (owing to bidirectional context) [12, 14, 15]. Within NAR frameworks, models can be broadly categorized into continuous-latent-based [14, 16, 17, 18] and discrete-token-based [19]. The latter was shown to yield superior prosodic diversity [20].

However, state-of-the-art (SOTA) discrete-token NAR systems [19] typically rely on complex two-stage cascaded pipelines (text-to-semantic followed by semantic-to-acoustic). While such decoupling simplifies the training of individual modules, it also introduces significant drawbacks: (1) error propagation, where inaccuracies in semantic prediction degrade the final audio quality; and (2) information bottlenecks, where low-bitrate semantic representations sacrifice fine-grained acoustic details. While single-stage alternatives [21] attempt to bypass these issues, they have historically lagged behind two-stage systems in terms of speech intelligibility.

[1] `https://github.com/k2-fsa/OmniVoice`

Preprint. Under review.

The main model components in this study are:

- **Input data:** Filipino text transcripts, Filipino audio, optional reference voice audio, and language ID `fil`.
- **Training target:** 8-codebook acoustic token sequences extracted from Filipino WAV files.
- **Learning method:** supervised fine-tuning of a pretrained neural TTS model.
- **Baseline:** original `k2-fsa/OmniVoice` checkpoint without Filipino-specific fine-tuning.
- **Evaluation:** matched comparison of generated audio from base and fine-tuned checkpoints.

4.4 Audio Tokenization

OmniVoice does not train directly on raw waveforms. Its data preparation flow converts each audio file into discrete audio tokens and stores those tokens in WebDataset shards. The official OmniVoice documentation states that audio files are resampled to 24 kHz and converted into 8-layer discrete tokens using the configured audio tokenizer. The default tokenizer path in the OmniVoice examples is `eustlb/higgs-audio-v2-tokenizer`, and the OmniVoice paper describes the Higgs Audio tokenizer as the component used to extract 8-codebook acoustic tokens and reconstruct audio (Zhu et al., 2026; Boson AI, 2025).

This tokenization step is central to the learning problem. A waveform is converted from a continuous signal into a matrix of acoustic token IDs. Each row corresponds to one codebook layer, and each time step represents a short audio frame. In OmniVoice, the model is trained to recover masked acoustic tokens conditioned on text tokens, language information, prompt audio tokens, and any unmasked target tokens. During inference, the model predicts target acoustic tokens through iterative unmasking, and the Higgs Audio decoder reconstructs waveform audio from the generated token matrix.

For this study, tokenization is performed once and reused as a Kaggle dataset artifact when possible. This avoids repeatedly spending GPU and disk time on audio token extraction during hyperparameter tuning.

4.5 Fine-Tuning and Experiment Tracking

Fine-tuning is performed in Kaggle using T4 GPU notebooks. Because the available GPU memory is limited, the setup uses FP16 precision, SDPA attention, smaller token batches, gradient accumulation, and checkpointing. ZeRO-style memory optimization is used when necessary to reduce optimizer-state memory pressure during training (Rajbhandari et al., 2019). Training starts from `k2-fsa/OmniVoice` rather than random initialization.

The fine-tuning objective follows OmniVoice's discrete masked diffusion formulation. Zhu et al. (2026) define the model as a single-stage non-autoregressive system that directly maps text and prompt-acoustic context to multi-codebook acoustic tokens. During training, a target acoustic-token segment is partially replaced with the mask token, and the model learns to recover the original token IDs at the masked positions. The loss is the negative log-likelihood of the masked target acoustic tokens, which is implemented in OmniVoice as cross-entropy over the audio-token vocabulary while ignoring non-loss positions. In this project, that means style tokens, text tokens, prompt audio, and unmasked target tokens are excluded from the loss with the ignore index, while the masked target audio tokens contribute to the update.

This objective is important for interpreting the training curves. The logged `train/loss` and `eval/loss` values are not waveform losses and are not directly WER, SIM-o, or UTMOS. They measure masked acoustic-token recovery under the same objective used by OmniVoice. The implementation computes per-codebook cross-entropy, averages each codebook's valid masked positions, and combines the 8 codebook losses using the configured weights `[8, 8, 6, 6, 4, 4, 2, 2]`. These larger weights on lower acoustic codebooks are consistent with the idea that earlier codebooks carry more coarse acoustic structure, while later codebooks add finer detail. The masking ratio is sampled from `[0.0, 1.0]`, matching the full-codebook random masking idea in OmniVoice, and the prompt ratio is sampled from `[0.0, 0.3]` so the model sees different amounts of reference acoustic context.

Optimization also follows the OmniVoice recipe but is scaled down for Kaggle. The OmniVoice paper reports AdamW with a cosine learning-rate schedule and warmup for large-scale training on H800 GPUs (Zhu et al., 2026). Our Kaggle runs used AdamW with weight decay 0.01, maximum gradient norm 1.0, seed 42, and cosine learning-rate decay after a 1% warmup. The learning rates were deliberately lower than OmniVoice's large-scale pretraining setting because this project updates an already pretrained checkpoint on a much smaller Filipino subset.

The comparison uses three controlled runs that share the same 5000-step budget and the same checkpoint policy and differ only in learning rate: `5e-6`, `1e-5`, and `2e-5`. With the 1% warmup, each run uses 50 warmup steps.

All three runs evaluated the development set at fixed intervals and saved a checkpoint only when development loss improved, so each run contributes its best-development-loss checkpoint to the comparison. In the `1e-5` run the lowest development loss, 4.162, occurred at step 4900, so checkpoint-4900 became that run's selected model; in the `2e-5` and `5e-6` runs the lowest development loss occurred at the final step, 5000. Differences between checkpoints are small in loss terms, which is exactly why downstream audio metrics, not loss alone, decide the final model in Section 5.

Weights and Biases is used for experiment tracking (Biewald, 2020). Each training run logs configuration values, training loss, development loss, checkpoints, generated samples, and evaluation artifacts, and each run's selected checkpoint is stored as a W&B artifact. The consolidated W&B report for this project is available as the OmniVoice PLD Filipino fine-tuning report. The logged runs make it possible to compare whether lower development loss corresponds to better WER, SIM-o, UTMOS, and listening quality. The run record of the recommended `1e-5` run and the generated-audio sample tables are reproduced in Appendices A and B.

Table 3
Fine-tuning setup.

Component	Configuration
Base model	`k2-fsa/OmniVoice`
Dataset	Clean Filipino PLD read speech
Training platform	Kaggle notebook
GPU	Kaggle T4 GPU environment
Precision	FP16
Attention	SDPA
Audio tokenizer	Higgs Audio V2 tokenizer
Logging	Weights and Biases
Tuning variable	Learning rate (5e-6, 1e-5, 2e-5)
Steps per controlled run	5000
Optimizer	AdamW
LR schedule	Cosine decay with 1% warmup
Loss	Weighted cross-entropy on masked acoustic-token positions
Checkpoint strategy	Best development-loss checkpoint per run

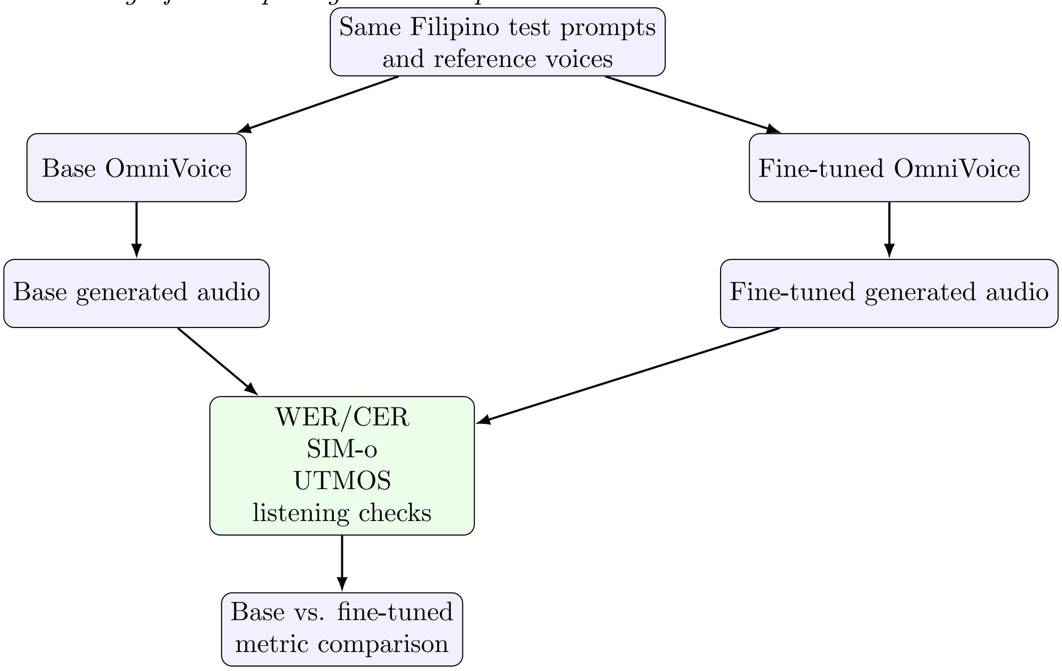
4.6 Evaluation Design

The evaluation compares the base OmniVoice model and Filipino fine-tuned checkpoints on the same test prompts and reference voices.

For each of the 4,322 test utterances, the model generates the target text while cloning a reference voice. The reference prompt is a different utterance from the same test speaker, taken as the next utterance from that speaker in the manifest, so the model never hears the audio it must reproduce. The same pairing is reused for every system. Inference uses 32 unmasking steps with batch size 2 on two T4 GPUs. The base model was measured in a first full evaluation run; later evaluation runs measured each best-development-loss fine-tuned checkpoint and reused the base row, which is valid because the test split, manifests, inference parameters, and evaluator versions were unchanged across runs.

Figure 4

Evaluation design for comparing model outputs.



4.7 Evaluation Metrics

The evaluation uses three objective metrics from modern zero-shot TTS evaluation: WER/CER for intelligibility, SIM-o for speaker similarity, and UTMOS for naturalness. Because each metric depends on a particular evaluator model or preprocessing pipeline, all evaluation runs use the same test split, inference settings, evaluator checkpoints, and text normalization before comparing base and fine-tuned results.

4.7.1 Word Error Rate and Character Error Rate

Word error rate (WER) measures whether the generated speech communicates the intended text. The generated waveform is first transcribed by an automatic speech recognition (ASR) model. The ASR transcript is then compared with the ground-truth target text using edit-distance operations: substitutions (S), deletions (D), and insertions (I). These edit operations trace back to Levenshtein distance (Levenshtein, 1966). WER is computed as:

$$\mathrm{WER} = \frac{S + D + I}{N} \times 100, \tag{1}$$

where N is the number of words in the reference text. Lower WER indicates that the ASR system recognized the generated speech as closer to the intended sentence. Character error rate (CER) follows the same idea but computes edit operations over characters instead of words, which can be useful for languages or text-normalization settings where word segmentation affects the metric.

In TTS evaluation, WER is not just a mathematical string distance; it is also dependent on the ASR model used to transcribe generated audio. The OmniVoice evaluation documentation reports different WER backends by benchmark: HuBERT-based WER for LibriSpeech-PC, Whisper for English Seed-TTS, Paraformer for Chinese Seed-TTS, Omnilingual-ASR for FLEURS, and Whisper or Paraformer for MiniMax multilingual evaluation depending on language (Zhu et al., 2026; Hsu et al., 2021; Radford et al., 2022; Gao et al., 2022). For this Filipino evaluation, the notebook uses the OmniVoice MiniMax multilingual WER path with Whisper-large-v3 for the non-Chinese Filipino setting and logs insertion, deletion, substitution, and total-word counts. This makes WER useful for comparing base and fine-tuned Filipino outputs, but the result should still be interpreted as ASR-mediated intelligibility rather than a direct human comprehension score.

4.7.2 Objective Speaker Similarity

SIM-o measures whether generated speech preserves the voice identity of the reference prompt. Voicebox popularized reporting objective similarity alongside listener-based speaker similarity for zero-shot TTS evaluation (Le et al., 2023). OmniVoice follows the same evaluation family and reports SIM-o as an objective speaker-similarity metric (Zhu et al., 2026).

Computationally, SIM-o is the cosine similarity between speaker embeddings extracted from the reference audio and generated audio:

$$\text{SIM-o}(r, g) = \frac{\mathbf{e}_r \cdot \mathbf{e}_g}{\|\mathbf{e}_r\| \|\mathbf{e}_g\|}, \tag{2}$$

where $\mathbf{e}_r$ is the speaker embedding of the reference prompt and $\mathbf{e}_g$ is the speaker embedding of the generated speech. Higher values indicate that the generated speech is closer to the reference speaker in the embedding space.

The reliability of SIM-o depends on the speaker encoder. The OmniVoice evaluation path uses WavLM-based ECAPA-TDNN speaker embeddings. ECAPA-TDNN is a speaker verification architecture that improves TDNN-style speaker embeddings with channel attention, propagation, and aggregation mechanisms (Desplanques et al., 2020). WavLM provides self-supervised speech representations that are useful beyond ASR, including speaker-related tasks (Chen et al., 2021). In this study, SIM-o is used to check whether Filipino fine-tuning improves intelligibility without damaging voice cloning identity. A fine-tuned model that reduces WER but lowers SIM-o substantially may be clearer but worse at preserving the reference speaker.

4.7.3 UTMOS

UTMOS estimates speech naturalness using a learned mean opinion score (MOS) prediction model. Saeki et al. (2022) introduced UTMOS as the UTokyo-SaruLab system for the VoiceMOS Challenge 2022, where the goal was to predict human MOS ratings for speech samples. Unlike WER, which evaluates textual intelligibility, and SIM-o, which evaluates speaker identity, UTMOS estimates whether the generated waveform sounds natural or human-like.

In the evaluation pipeline, each generated audio file is passed to the UTMOS model, and the resulting predicted MOS values are averaged for each system. Higher UTMOS indicates better predicted naturalness. However, UTMOS is still a predictor trained from listening-test data, so it should not be treated as a complete replacement for human listening checks. For this reason, UTMOS is interpreted together with WER, SIM-o, and selected Filipino listening samples. For example, a model may obtain a high UTMOS score because the waveform is smooth while still mispronouncing Filipino words; conversely, a model may improve WER while sounding less natural.

4.7.4 Metric Reliability

The most reliable comparison is a controlled within-study comparison rather than an absolute claim of model quality. Both base and fine-tuned models are evaluated on the same Filipino test prompts, reference voices, inference parameters, ASR model, speaker encoder, UTMOS checkpoint, and text normalization rules. Weights and Biases stores the metric summaries, raw metric logs, generated audio examples, and checkpoint metadata. This makes it possible to compare hyperparameter runs by both development loss and downstream audio metrics.

5 Results and Discussion

5.1 Objective Metric Comparison

Table 4 reports the full test-set results for the base model and the three controlled best-development-loss checkpoints, Figure 5 plots the WER comparison, and Figure 6 shows all three metrics side by side.

Table 4

Full PLD Filipino test-set objective results for the controlled comparison. All fine-tuned rows are the best-development-loss checkpoints of 5000-step runs that differ only in learning rate. Lower WER is better; higher SIM-o and UTMOS are better. Bold marks the best value per column.

Model	Selection	WER (%)	ΔWER	SIM-o	UTMOS
Base OmniVoice	pretrained base	22.55	0.00	0.602	**3.64**
Fine-tuned, lr 5e-6	best dev loss at step 5000	21.96	-0.59	**0.605**	3.60
Fine-tuned, lr 1e-5	best dev loss at step 4900	**18.52**	-4.03	0.604	3.61
Fine-tuned, lr 2e-5	best dev loss at step 5000	18.83	-3.72	0.583	3.61

Figure 5

Test-set WER by system. The dashed line marks the base-model WER; all three fine-tuned checkpoints fall below it, with the 1e-5 and 2e-5 checkpoints clearly ahead.

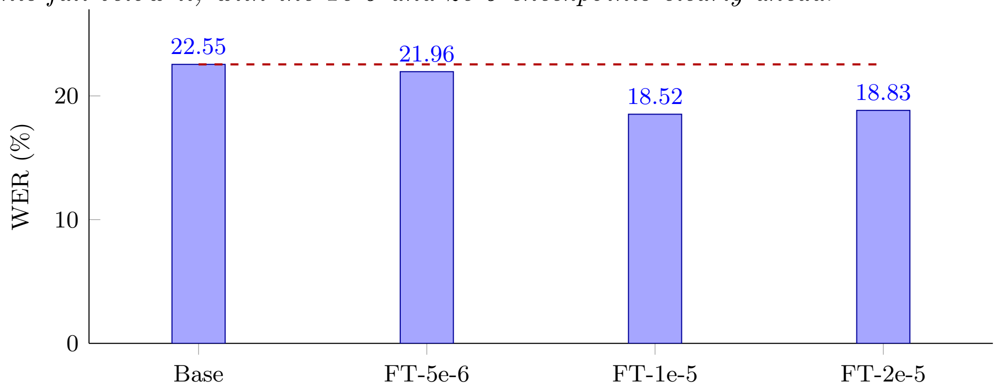
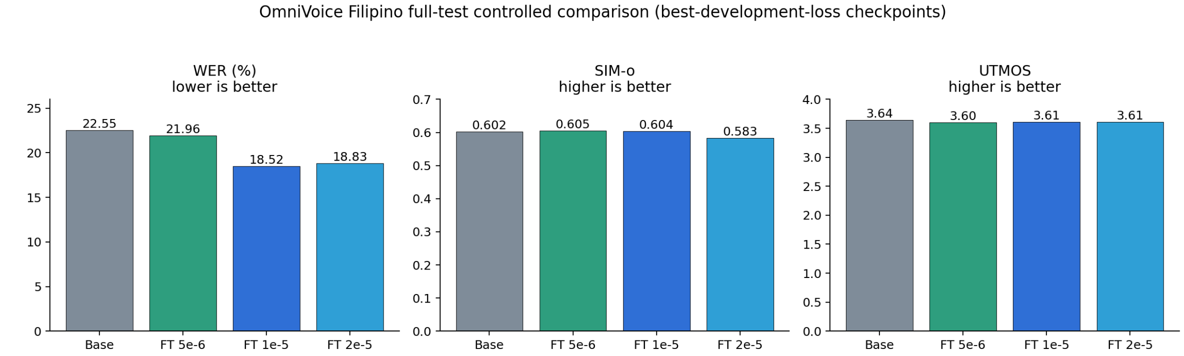


Figure 6

WER, SIM-o, and UTMOS for the final controlled comparison on the full PLD Filipino test split.



Because the fine-tuned rows share the same step budget and checkpoint policy, the metric differences in Table 4 are attributable mostly to learning rate, and they form a consistent pattern. The 1e-5 checkpoint is the strongest overall: it reduced WER to 18.52%, a 4.03 absolute point improvement over the base model and about a 17.9% relative WER reduction, while its SIM-o of 0.604 stays slightly above the base model's 0.602. The 2e-5 checkpoint nearly matches that intelligibility gain at 18.83% WER, but it has the lowest SIM-o of all four systems at 0.583, the only fine-tuned value below the base model. The 5e-6 checkpoint moves least: it has the best fine-tuned SIM-o at 0.605 but improves WER by only 0.59 points.

UTMOS separates the systems least. The base model keeps the highest predicted naturalness at 3.64, and the three fine-tuned checkpoints sit close together at 3.60–3.61. These UTMOS and SIM-o differences are small and should not be over-interpreted, but their direction is consistent: fine-tuning toward Filipino intelligibility costs a small amount of predicted naturalness, and pushing the learning rate to 2e-5 additionally costs speaker similarity. The comparison is therefore an intelligibility-versus-speaker-similarity tradeoff governed by learning rate rather than a single-metric win. The 1e-5 checkpoint is the recommended model because it captures nearly all of the available intelligibility gain without giving up speaker similarity.

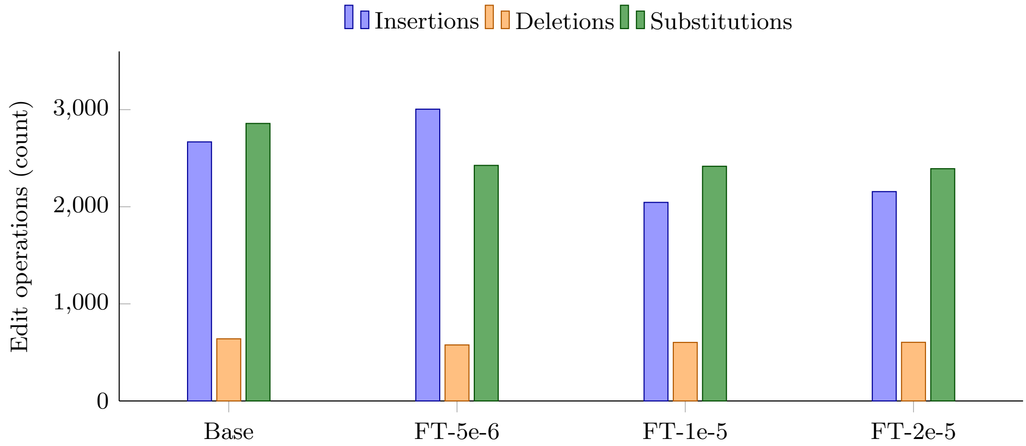
5.2 WER Error Analysis

Table 5

WER edit-operation counts on the full PLD Filipino test set for the controlled comparison.

Model	Insertions	Deletions	Substitutions	Words
Base OmniVoice	2,668	640	2,858	27,342
Fine-tuned, lr 5e-6	3,005	577	2,426	27,354
Fine-tuned, lr 1e-5	2,045	603	2,417	27,354
Fine-tuned, lr 2e-5	2,156	604	2,392	27,354

Figure 7
Edit operations per system. The 1e-5 and 2e-5 checkpoints win mainly by cutting insertions and substitutions; the 5e-6 checkpoint loses most of its substitution gain to an insertion increase.



The edit-operation counts in Table 5 and Figure 7 explain the model ranking. The 1e-5 checkpoint reduces insertions from 2,668 to 2,045 and substitutions from 2,858 to 2,417 relative to the base model, while deletions stay near the base level; its improvement comes from producing fewer extra words and fewer wrongly recognized words. The 2e-5 checkpoint follows the same pattern with slightly more insertions and slightly fewer substitutions. The 5e-6 checkpoint instead cuts substitutions to 2,426 but raises insertions to 3,005, above the base model, and that single increase absorbs most of its substitution gain. A fine-tuned model can improve one error type while worsening another, which is why aggregate WER should be read together with its components.

5.3 Development Loss versus Downstream Metrics

Development loss was used to monitor training and select each run's checkpoint, and its relationship to downstream quality turned out to be useful but incomplete. Selecting checkpoints by development loss gave every run a principled stopping point, and the selected 1e-5 checkpoint did produce the lowest WER of the study. But development loss could not have predicted the full metric pattern: the 1e-5 and 2e-5 checkpoints are separated by only 0.31 WER points yet by 0.021 SIM-o, and the 5e-6 checkpoint's insertion problem is invisible in its loss curve. Development loss is a good checkpoint-selection heuristic within a run, but checkpoints across runs must be compared with downstream audio metrics before declaring a winner.

The appendix screenshots preserve the W&B evidence used to inspect training behavior and generated samples. Appendix A shows the run record of the recommended `1e-5` run, and Appendix B shows the generated-test-sample tables.

5.4 Limitations

The completed comparison uses automatic metrics only. WER depends on the ASR model (Whisper-large-v3 decoding Tagalog), so it estimates intelligibility as that ASR system hears it, and ASR for Filipino is itself imperfect; SIM-o depends on the speaker encoder; and UTMOS is a learned MOS predictor rather than a listening survey, so the small naturalness differences reported here are predictions, not human judgments. No human listener study has been conducted. The corpus is Filipino read speech with many short, isolated-word prompts, so the results say little about spontaneous speech, long-form prosody, or other Philippine languages.

Kaggle T4 compute limits also restricted the number of hyperparameter runs, so the controlled learning-rate comparison samples three points rather than a full sweep, with one run per learning rate and no variance estimate across seeds.

6 Conclusion and Future Work

Supervised Filipino fine-tuning improved OmniVoice intelligibility on the Philippine Languages Database test set. In the controlled comparison of best-development-loss checkpoints from identical 5000-step runs, the 1e-5 checkpoint reduced WER from 22.55% to 18.52% while keeping SIM-o slightly above the base model; the 2e-5 checkpoint reached a similar 18.83% WER but lowered SIM-o below base; and the 5e-6 checkpoint preserved speaker similarity best while improving WER only modestly. The base model retained the highest predicted naturalness. The learning rate therefore controls an intelligibility-versus-speaker-similarity tradeoff, and fine-tuned models must be compared with the full downstream metric pattern rather than training loss, development loss, or any single metric alone.

Future work should add a Filipino listener evaluation to validate the automatic metrics, sample more learning rates around 1e-5, repeat runs across seeds to estimate variance, and build a verified subset of longer sentence-level utterances to test whether sentence-level training data improves Filipino prosody. The recommended checkpoint for demonstration and reporting is the best-development-loss checkpoint of the 1e-5 run, selected at step 4900.

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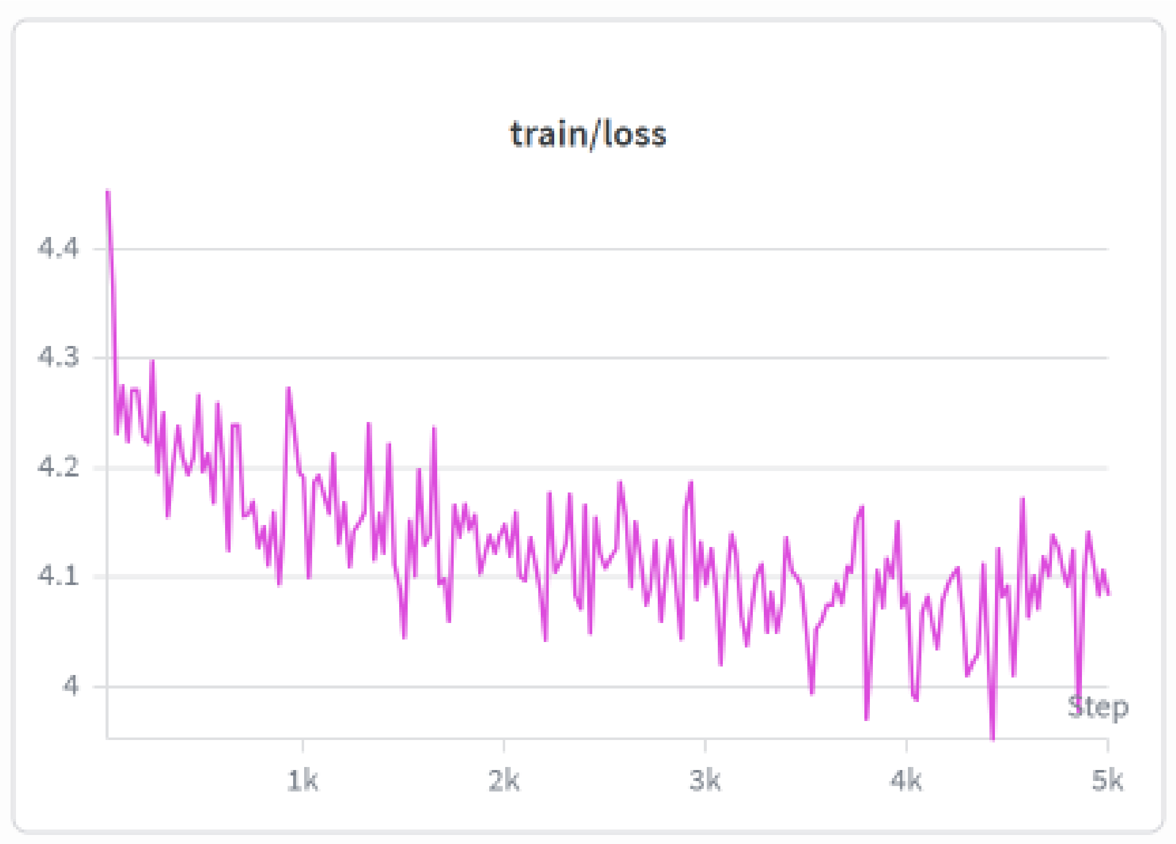
Appendix A
Fine-Tuning Run Record

The full interactive experiment report is available as the OmniVoice PLD Filipino W&B report.

Figure A1

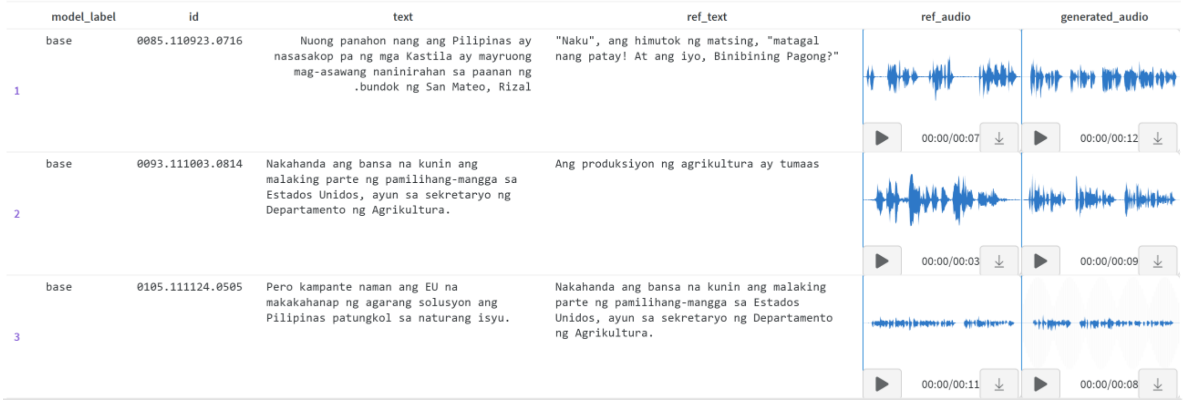
W&B run screenshot for the 5000-step fine-tuning run at learning rate 1e-5, whose best-development-loss checkpoint (step 4900) is the recommended model.



Appendix B
Generated Test Sample Tables

Figure B1

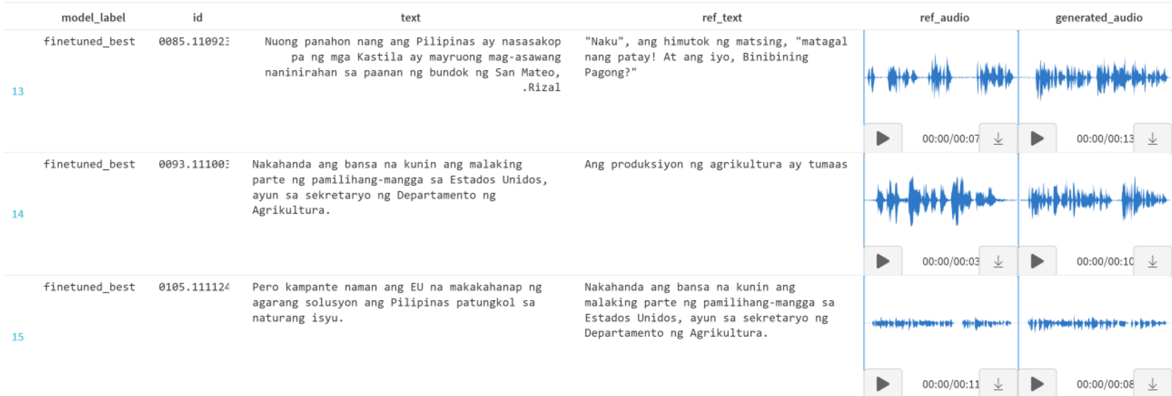
W&B generated-test-samples table for the base OmniVoice model.



	model_label	id	text	ref_text	ref_audio	generated_audio
1	base	0085.110923.0716	Nuong panahon nang ang Pilipinas ay nasasakop pa ng mga Kastila ay mayruong mag-asawang naninirahan sa paanan ng .bundok ng San Mateo, Rizal	"Naku", ang himutok ng matsing, "matagal nang patay! At ang iyo, Binibining Pagong?"	00:00/00:07	00:00/00:12
2	base	0093.111003.0814	Nakahanda ang bansa na kunin ang malaking parte ng pamilihang-mangga sa Estados Unidos, ayun sa sekretaryo ng Departamento ng Agrikultura.	Ang produksiyon ng agrikultura ay tumaas	00:00/00:03	00:00/00:09
3	base	0105.111124.0505	Pero kampante naman ang EU na makakahanap ng agarang solusyon ang Pilipinas patungkol sa naturang isyu.	Nakahanda ang bansa na kunin ang malaking parte ng pamilihang-mangga sa Estados Unidos, ayun sa sekretaryo ng Departamento ng Agrikultura.	00:00/00:11	00:00/00:08

Figure B2

W&B generated-test-samples table for the 5000-step learning-rate 1e-5 fine-tuning run.



	model_label	id	text	ref_text	ref_audio	generated_audio
13	finetuned_best	0085.11092	Nuong panahon nang ang Pilipinas ay nasasakop pa ng mga Kastila ay mayruong mag-asawang naninirahan sa paanan ng bundok ng San Mateo, .Rizal	"Naku", ang himutok ng matsing, "matagal nang patay! At ang iyo, Binibining Pagong?"	00:00/00:07	00:00/00:13
14	finetuned_best	0093.11100	Nakahanda ang bansa na kunin ang malaking parte ng pamilihang-mangga sa Estados Unidos, ayun sa sekretaryo ng Departamento ng Agrikultura.	Ang produksiyon ng agrikultura ay tumaas	00:00/00:03	00:00/00:10
15	finetuned_best	0105.11112	Pero kampante naman ang EU na makakahanap ng agarang solusyon ang Pilipinas patungkol sa naturang isyu.	Nakahanda ang bansa na kunin ang malaking parte ng pamilihang-mangga sa Estados Unidos, ayun sa sekretaryo ng Departamento ng Agrikultura.	00:00/00:11	00:00/00:08